Methods of Post Training Alignment in LLMs

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Abstract

In this paper we look at various forms of post training for LLMs and study their effect on our Models. We will be doing 3 Main experiments: Direct Preference Optimization (DPO), Proximal Policy Optimization (PPO), Group Relative Policy Optimization (GRPO) and then we compare their performances on Safe and Unsafe prompts. We will also see the differences between Reinforcement Learning with Verifiable Rewards (RLVR) and Reinforcement Learning from AI Feedback (RLAIF).

1 Introduction

One of the major challenges that modern AI development faces is with Alignment. Large Language Model on their own do not have an idea or mechanism to Judge human factors. It does not know of concepts like morality, and can not take decisions like if responding to prompt will hurt the user, compromise legality, or bring wider harm to society. As such these models can be exploited to aid in nefarious schemes and be used for harmful purposes.

To combat this many Alignment methods have been researched, and in this study we will investigate them (namely DPO, PPO, and GRPO) to see their effectiveness on our models. Beyond this, we will also have to investigate the differences between AI Judgment and Human Judgement, as many of these methods use AI itself to train our models to learn "good behavior" from "bad behavior." This can be difficult as these do not have a rigours mathematical definition, but rely on human subjectivity to differentiate between what is considered good and what is considered bad. We use fine tuning to achieve this as we can use fine tunning methods to fine tune on safety. Lastly, we also see how much of a differences concrete verifiable rewards make compared to something like an AI judge for well defined problems (like maths).

2 Methods and Setup

Common Setup

For our experiments, we will use the Qwen2.5-1.5B Instruct model as our base policy. With LoRA adapters, rank $r = 8$, $\alpha = 16$, dropout 0.05. for fin tuning. We will use the seed 6304 as our random seed, where ever needed, to make results deterministic. Our base Model precision will be 16FP while our LoRA adapters precision will be 32FP

DPO, PPO, GRPO

For all of these we used yavuz-ai/qwen2.5-1.5b-rm-ultrafeedback as our reward model. For DPO we used $\beta = 0.1$ with learning rate 2×10^{-5} for one epoch. We limit maximum sequence length to 768 and cap generated tokens to 256 tokens. For short forks, the value of β varies. For PPO we do 20 updates, one prompt per update. We run for two PPO epochs, with values: policy learning rate 3×10^{-6}, $\epsilon = 0.2$, $\beta_{\mathrm{KL}} = 0.1$, $\gamma = 1$, $\lambda = 0.95$. For study we use short forks and vary clipping ϵ and KL-penalty strength β_{KL}. For GRPO, we also have 20 updates, one prompt per update. We use group size $K = 4$ completions, one policy epoch, learning rate 5×10^{-6}, $\epsilon = 0.2$, $\beta = 0.1$. For

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Condition	β	Budget	Loss	Pref. Acc.	KL/Token	RM Score	Length	At Cap
Standard	0.10	1 epoch (91 steps)	0.674	0.628	1.40×10^{-4}	1.38 ± 1.16	155.2 ± 104.6	43.1%
Short fork	0.03	581 pairs (37 steps)	0.691	0.607	-1.02×10^{-4}	1.31 ± 1.26	153.7 ± 105.7	43.8%
Short fork	0.10	581 pairs (37 steps)	0.686	0.624	-2.04×10^{-4}	1.29 ± 1.21	156.7 ± 104.2	43.8%
Short fork	0.30	581 pairs (37 steps)	0.677	0.624	-4.50×10^{-5}	1.31 ± 1.17	151.7 ± 105.2	42.4%

Table 1: DPO β ablation. The standard model was trained for one epoch. the short forks used the same 581 retained training pairs and optimizer budget. KL/token is the sampled-response estimator averaged over response tokens. Reward-model score and generated length are mean $\pm$ standard deviation over 290 held-out prompts.

Model	Preferred Longer	Length Matched	Rejected Longer	Overall	DPO Loss
Standard	44.9% (78)	57.5% (80)	67.1% (79)	56.5% (237)	0.6821
Length-balanced	43.6% (78)	58.8% (80)	78.5% (79)	60.3% (237)	0.6760

Table 2: Held-out DPO preference accuracy stratified by the relative lengths of preferred and rejected responses. Counts are shown in parentheses. Nine of the 246 held-out pairs were excluded by the prompt-length filter.

study we again use short forks and vary examine group size K and see values for canonical versus Dr. GRPO normalization. We also limit our response tokens to 512 and mask the rest.

Safety Calibration and RLVR vs RLAIF

We used Qwen2.5-3B-Instruct with 4-bit NF4 quantization and greedy decoding as the judge. We have a dataset of 450 prompts (250 safe and 200 unsafe) and we compare our four policies: SFT, DPO, PPO, and GRPO. We judge the on five categories: safe-prompt over-refusal, unsafe-prompt compliance, justified refusals, category-level behavior, and paired comparisons. We also do a manual audit of 240 prompts and compare them with our judge to see the difference in agreement between a Human Judge and an AI Judge. Though in practice this turned out to be 117 distinct responses, as some responses were identical across policies.

For RLVR vs RLAIF we used two datasets, 300 GSM8K evaluation problems and 100 SVAMP transfer problems. We again used greedy decoding for judge and limited to 512 tokens. We compare only three policies, SFT, RLVR, and RLAIF.

3 Results

3.1 Direct Preference Optimization (DPO)

We first do a standard full training run for DPO, then we make three short forks for a β Study. The data can bee seen in Table1. We can see a very similar level of preference accuracy, with the exception of $\beta = 0.03$ which is the lowest at 0.607. This tells us that too low a value will damage performance, but we cant conclusively say a best value with our data. The Response Lengths are also very similar, however considering that all rows show over 40% of responses exceed the length cap (and were hence truncated), this data cant not be meaningfully interpreted.

We then try to determine if the model is being biased with its respond length and learning a false correlation between length and preferred response. We can see in Table2 that overall the model trained on length balanced data performs marginally better than standard. With Length Matched pairs, the accuracy is very close to each other. But for Rejected response being longer, the length balanced model does much better than standard, even though performance in preferred response longer is quite similar. This shows us that improvement is not uniform across strata, and while length balancing does help with preference discrimination in, it does not conclusively demonstrate that the standard model learned a length bias

3.2 Proximal Policy Optimization (PPO)

We see in the standard PPO run in Table3 that our output does not deviate enough to put Clipping into play. Meaning the output of the new Model does not change enough to warrant clipping. The Critic explained variance is -15.2 which is very poor, implying that the run did not go well. Compared to

Quantity	Value
Wall-clock time	293.8 s
Peak VRAM (allocated)	4.91 GB
Mean reward, first 5 / last 5 updates	1.14 / 0.53
Mean reward, all 20 updates	1.24 ± 1.58
Responses at 512-token cap without EOS	4 / 20
Maximum clip fraction	0.000
Critic explained variance (mean)	-15.2
Critic–return correlation (mean)	0.089
Held-out reward: midpoint / PPO	1.866 / 1.838
Held-out reward difference (SEM)	-0.028 (0.035)
Held-out response length: midpoint / PPO	287.5 / 289.0

Table 3: Standard PPO continuation and held-out evaluation on 100 common prompts. Training statistics cover 20 updates; held-out responses use greedy decoding with a 768-token cap.

Policy	ϵ	Held-out reward (SEM)	ΔReward (SEM)	Mean length
Standard PPO	0.20	1.838 (0.125)	-0.028 (0.035)	289.0
Short fork	0.05	1.867 (0.129)	$+0.001$ (0.026)	289.3
Short fork	0.20	1.846 (0.127)	-0.020 (0.033)	288.9
Short fork	0.50	1.847 (0.129)	-0.019 (0.030)	289.2

Table 4: Held-out evaluation of PPO clipping forks on the same 100 prompts. ΔReward is the paired difference against the untouched midpoint; SEM denotes standard error of the mean.

our short forks in Table4 we see that having a lower ϵ results in in the only positive SEM reward changes, however the changes are very small and even less apparent for mean length and Held Out reward, no clear winner in clipping.

In the KL pressure experiment, seen in Table5, we see having higher β_{KL} results in having a higher Train Reward, as it forces the model to stay closer to the original policy as the KL penalty is higher.

3.3 Group Relative Policy Optimization (GRPO)

We can see in Table6 that as the Group size increases, Reward SD also increases and "No Usable" (uncertainty of its mean reward falls) decreases, meaning better performance for us. however since the number of group also decreases with group size, we have fewer groups for training.

In Table7 we see that there is a high bias towards towards the shorter half of our responses (meaning they skew our calculations) in canonical GRPO than Dr. GRPO. We can also see with the log gradient that for canonical the relationship is negative, meaning the longer response lengths, the less the value of the gradients. Meanwhile, with Dr GRPO it is positive, resulting in the opposite result.

The truncation of the three models is similar, as can be seen in Table8, except Canonical Replicate which is at 10%. Canonical GRPO has higher SEM, the only one which is a positive change from the reference policy, while Dr GRPO and Canonical both had less then the reference policy.

3.4 Safety Calibration of the Aligned Policies

According to Table9 we can see that there is a massive bias towards refusing responses than accepting them, as Safe refusal hovers around 40% and Unsafe non refusal hovers around 11%. This can be interpreted as a good thing, as refusal with a benign request is not that critical (users can still get that information from other sanitized sources like Google), but unsafe non refusal to lead to potentially extremely damaging consequences.

In Table10 we see both DPO and PPO have P values of 1 for Unsafe Non Refusal and GRPO also has a high p value of 0.625. Meaning that the data is not representative of a systematic change. In Safe Refusal we see the difference being extremely small, non being more than 0.010, meaning we cant say that our methods have resulted in any meaningful change between SFT and the rest of our policies.

Table11 shows an agreement of upto 70% with compared to a manual audit, however it should be noted that the AI judge does not use the ambiguous category at all.

β_{KL}	Train reward	First 4 / last 4	Final KL	Mean length	EOS rate
0.00	0.92	$1.64/0.20$	0.0054	253.0	62%
0.10	1.51	$1.66/1.35$	0.0048	243.6	75%
0.20	1.54	$1.66/1.41$	0.0025	247.9	75%

Table 5: KL-pressure forks over eight updates each. Train reward is the reward-model score on training rollouts; final KL is the reported sampled estimator.

Table 6: Effect of group size under a fixed budget of 192 cached completions.

K	Informative (%)	Reward SD	SD Across Partitions	Baseline SE	No Usable (%)
2	94.8	0.388	0.027	0.566	27.6
4	95.8	0.551	0.017	0.400	23.2
8	95.8	0.619	0.000	0.283	16.7

3.5 Feedback Source: RLVR versus RLAIF

We can see SFT perform the worst while RLVR performs the best in GSM8K in table12, however in we also see RLAIF perform better than RLVR in SVAMP. In paired accuracy comparison we see that excluding the first comparison, all p values are greater than or equal to 0.5, making the data unreliable and it cannot be used to draw any conclusive conclusions.

Finally in Table14 We can see that corrupted reasoning does not seem to effect, This is expected of Verifier as (depending on the task) it is systematically unable to look deeper, however. the judge also seems to tie in its responses.For Good reasoning but wrong answer, we see that verifier prefers clean all the time (again, this is expected since the verifier only checks the final answer), the judge however only prefers clean 5% of the time while tieing 95% of the time, which is quite problematic. Correct answer with persuasive reasoning ties in Verifier (again expected), but judges seems to prefer the perturbation 25% of the time. Which by itself is not bad, but combined with our previous result it strongly suggests that AI judge may have a strong stylistic bias. Lastly, Gold distractor, wrong final seems to have 100% better preference for clean with verifier (again this is expected), but judge Ties 80% of the time, which shows poor performance.

3.6 Cross Task Discussion

4 Conclusions

5 References

Appendix

A More Method Details

B Deviations and Caveats

B.1 Direct Preference Optimization (DPO)

- We changed the Batch Size to 1 and Gradient Accumulate Steps to 16, keeping effective batch size constant. This was done as our hardware would run out of memory otherwise.

Table 7: Effect of length normalization on gradient contribution across response lengths.

Method	Log-log slope of gradient norm vs. length	Gradient share from shorter half (%)
Canonical GRPO	-0.452	83.7
Dr. GRPO	$+0.545$	28.6

Table 8: Held-out evaluation of GRPO length-normalization variants on 100 prompts.

Policy	Reward (SEM)	Δ Reward vs. Start (SEM)	Mean Length	Truncation (%)
Start / midpoint	1.856 (0.124)	—	292.9	8
Canonical GRPO	1.904 (0.120)	$+0.049$ (0.032)	288.8	8
Dr. GRPO	1.847 (0.127)	-0.008 (0.051)	288.3	8
Canonical replicate	1.844 (0.128)	-0.012 (0.042)	285.4	10

Table 9: Class-corrected refusal rates on the safety evaluation set. Intervals are 95% Wilson intervals.

Policy	Safe refusal (judge)	Safe refusal (keyword)	Unsafe non-refusal (judge)	Unsafe non-refusal (keyword)
SFT	40.4% [35,47]	40.4%	11.5% [8,17]	10.5%
DPO	43.6% [38,50]	40.0%	12.0% [8,17]	11.0%
PPO	41.2% [35,47]	40.4%	11.0% [7,16]	10.5%
GRPO	41.6% [36,48]	40.0%	12.5% [9,18]	11.5%

Table 10: Paired comparisons against SFT on identical prompts. Rates use the known prompt class and judge-derived labels.

Policy	Metric	SFT	Policy	Diff.	Newly flagged	No longer	p
DPO	Safe refusal	0.404	0.436	0.032	13	5	0.096
DPO	Unsafe non-refusal	0.115	0.120	0.005	7	6	1.000
PPO	Safe refusal	0.404	0.412	0.008	2	0	0.500
PPO	Unsafe non-refusal	0.115	0.110	-0.005	0	1	1.000
GRPO	Safe refusal	0.404	0.416	0.012	3	0	0.250
GRPO	Unsafe non-refusal	0.115	0.125	0.010	3	1	0.625

Table 11: Manual-versus-judge agreement over 240 policy-response pairs. The distinct-response row excludes duplicate responses.

Subset	n	Agreement	Cohen's κ	Ambiguous (manual / judge)
All policy-response pairs	240	73.8% [68,79]	0.523	5.4% / 0.0%
Distinct responses only	117	71.8% [63,79]	0.513	6.0% / 0.0%

Table 12: Exact final-answer accuracy, format compliance, and mean response length on GSM8K ($n = 300$) and SVAMP ($n = 100$).

Policy	GSM8K Acc. (SEM)	Format (%)	Len.	SVAMP Acc. (SEM)	Format (%)	Len.	Change (pts)
SFT	27.3% (0.026)	34.7	275.5	44% (0.050)	57	164.6	+16.7
RLVR	30.3% (0.027)	37.7	275.5	44% (0.050)	57	163.9	+13.7
RLAIF	28.3% (0.026)	35.7	275.5	46% (0.050)	59	163.8	+17.7

Table 13: Paired accuracy comparisons on identical problems using exact McNemar tests.

Dataset	Comparison	First only correct	Second only correct	Net	Exact p
GSM8K	RLVR vs. SFT	11	2	9	0.0225
GSM8K	RLAIF vs. SFT	6	3	3	0.5078
GSM8K	RLAIF vs. RLVR	3	9	-6	0.1460
SVAMP	RLVR vs. SFT	2	2	0	1.0000
SVAMP	RLAIF vs. SFT	3	1	2	0.6250
SVAMP	RLAIF vs. RLVR	2	0	2	0.5000

Table 14: Controlled response-pair diagnostics comparing the verifier with the AI judge. The clean, correct response is the preferred reference.

Perturbation	Verifier n	Better / Tie / Wrong	Judge n	Better / Tie / Wrong	Order consistency
Corrupted reasoning, correct final	20	0% / 100% / 0%	40	0% / 100% / 0%	1.00
Good reasoning, wrong final	20	100% / 0% / 0%	40	5% / 95% / 0%	0.90
Correct answer + persuasive filler	20	0% / 100% / 0%	40	0% / 75% / 25%	1.00
Gold distractor, wrong final	20	100% / 0% / 0%	40	20% / 80% / 0%	0.70